NLP / LLM Document Understanding

Internship Assignment

This assignment evaluates how you approach real-world document understanding problems: page classification, multi-page reasoning, structured extraction, loan-level reconciliation, and AI-assisted engineering. **We care more about clear reasoning, verification, and engineering judgment than about a production-perfect solution.**

Timeline

You have 7 calendar days from the time you receive the assignment. This is not a speed test; use the time in a way that helps you produce a thoughtful and well-verified solution.

AI tools are allowed and encouraged

You may use coding agents, LLMs, IDE assistants, MCP servers, custom rules, skills, or any other tools you normally use. You remain responsible for the correctness of the final submission. We are interested in how you use AI as an engineering tool, not in whether you use a specific product or feature.

1. Provided Inputs

- AREAL_LOAN.pdf - 12 sequential pages from a mortgage loan package
- candidate_classification_labels.txt - allowed page/document labels
- extraction_fields.json - fields to extract at loan level

2. Required Deliverables

- **Written report (PDF or Google Doc):** Design decisions, prompts, reasoning, and concise discussion of the tasks below.
- **result.json:** Page labels, document grouping, final extracted loan-level values, and source page references.
- **Implementation (GitHub repo or ZIP):** A small implementation of at least one substantive part of your approach, plus a short README.
- **AI_WORKFLOW.md:** A short summary of your AI tools/agents, rules or skills, and verification workflow.
- **5-8 minute workflow recording:** A screen walkthrough of the actual development workflow and final solution. No slides or polished editing required.

Keep it concise

There is no minimum page count. We prefer a clear, evidence-based submission over a long report. Do not spend time making slides or over-polishing the presentation.

3. Tasks

Task 1 - Classify the Provided Loan

Classify every page in AREAL_LOAN.pdf using one of the labels in candidate_classification_labels.txt.

- Provide a page-level label for all 12 pages.
- Group pages that belong to the same logical document where appropriate.
- Store the result in a structured representation of your choice and briefly explain why you chose it.
- If a page is ambiguous, continuation-only, or difficult to classify, make that uncertainty visible rather than hiding it.

Task 2 - Classification Prompt Design

Design an LLM prompt that can perform the page classification task robustly.

- Provide the full prompt (system + user, or the equivalent structure used by your model/harness).
- Define the expected structured output format.
- Explain how the prompt handles continuation pages and multi-page context.
- Explain what you would do when the model is uncertain or when adjacent pages provide important context.

Task 3 - Multi-Page Processing & Extraction

Design the extraction stage for the fields in extraction_fields.json and apply it to the provided loan.

- Explain whether you would process pages individually, as grouped documents, or in another structure - and why.
- Provide the extraction prompt you would use and the expected schema.
- Explain when extraction is triggered (for example, after classification or only for relevant document types).
- Return your final loan-level extracted values in result.json and include the page(s) that support each value.
- Discuss relevant trade-offs such as context, accuracy, cost, consistency, and failure handling.

Task 4 - Minimal Implementation

Implement at least one substantive part of your approach. The goal is not production readiness; we want to see how you turn your design into working code.

- Examples: classification prompt runner, extraction pipeline, document grouping logic, triangulation/reconciliation logic, or an evaluation/validation layer.
- Your code should meaningfully interact with the assignment data or with the structured outputs you produce.
- Include a README explaining how to run the relevant part of the solution.
- If a paid API or unavailable credential prevents a fully runnable LLM call, keep the interface clear and explain how you validated the component you were able to run.

Task 5 - Loan-Level Triangulation

Mortgage packages often contain the same field in multiple documents. Your final output, however, must contain one consistent value for each loan-level field.

- Identify at least one real ambiguity, inconsistency, or competing observation in the provided loan.

- Explain how your system would reconcile that situation and why the selected value should be trusted.
- Discuss relevant signals such as document type reliability, source-page evidence, normalization, frequency, model confidence, and cross-document consistency.
- Apply your strategy to the provided loan; do not answer only with generic heuristics.

Task 6 - Agentic Development Workflow

We actively use AI coding agents in our engineering workflow. We therefore want to understand how you used them while completing this assignment.

AI_WORKFLOW.md (required, keep it short)

- Tools/harnesses used (for example: Codex, Claude Code, Cursor, Copilot, Gemini CLI, or others).
- Any repository-level instructions or agent configuration you used (for example: AGENTS.md, CLAUDE.md, .cursor/rules, SKILL.md/skill.md files, custom commands, MCP servers).
- How you broke the assignment into tasks and decided what to delegate to the agent.
- How you reviewed or verified generated code and outputs.
- One concrete example where the agent was wrong, incomplete, or headed in the wrong direction - and what you changed.

5-8 minute screen recording (required, in English)

- No slide deck is needed. A simple screen recording is enough; camera is optional.
- Show the actual repository/project structure and the final output.
- Show the AI tool/harness you used and representative visible interactions or prompts.
- Show any rules, SKILL.md/skill.md files, instruction files, custom commands, or MCP setup you actually used.
- Walk through one representative development task: how you gave context, what the agent produced, and how you reviewed/changed it.
- Show how you verified the result (tests, diffs, manual checks, output validation, or similar).

Important: judgment matters more than feature count

You will not receive extra credit simply for using more agents, skills, MCP servers, or custom configuration. If you did not use one of these features, say so. We care about deliberate tool use, verification, and your ability to stay in control of the engineering process.

Privacy / security

Do not include API keys, passwords, customer data, or unrelated private information in your repository, report, or recording. You do not need to reveal hidden chain-of-thought or private model reasoning; visible prompts, interactions, artifacts, and your own explanation are sufficient.

4. Suggested result.json Structure

You may use another structure if you have a good reason, but your output should contain equivalent information.

```
{
  "page_labels": [
    {"page_number": 1, "label": "<allowed label>", "confidence": null, "notes": ""}
  ],
  "documents": [
    {"label": "<document label>", "pages": [1, 2]}
  ],
  "loan_level_fields": {
    "borrower_name": {"value": "", "source_pages": []},
    "property_address": {"value": "", "source_pages": []},
    "loan_number": {"value": "", "source_pages": []}
  }
}
```

5. What We Evaluate

- Correctness and evidence on the provided loan
- Quality and robustness of prompt design
- Handling of multi-page context, ambiguity, and conflicting information
- Engineering reasoning and clarity of data structures
- Implementation quality, verification, and failure handling
- Maturity of AI-agent usage: context, delegation, review, iteration, and control
- Ability to communicate technical decisions clearly and concisely

6. Submission Checklist

- Written report (PDF or Google Doc)
- result.json
- Code repository link or ZIP + README
- AI_WORKFLOW.md
- 5-8 minute workflow recording link (in English)

Deadline: 7 calendar days after receiving the assignment.

Good luck - we are looking forward to seeing how you approach the problem.